

INTERACTIVE LAYER

Metadata GUI

metadata_gui

Search · edit · visualize unified metadata

Submission GUI

submission_gui

Guided metadata entry · GenBank packaging

metadata in

assists module 5

PIPELINE LAYER · CORE WORKFLOW

MODULE 1

Public Data Acquisition

public_metadata_pipeline

- SRA / GSA dual search
- AI-assisted query terms
- Metadata retrieval
- Batch download & convert
- Host genome DB build

→ FASTQs + metadata

MODULE 2

Novel Virus Discovery

virome_discovery_pipeline

- QC & host depletion
- De novo assembly
- 10-tool identification
- Taxonomy voting (8-way)
- Host prediction
- Rescue & report

→ contigs + taxonomy

MODULE 3

Known-Virus Analysis

virome_analysis_pipeline

- Quantification (TPM/RPM)
- Variant calling
- snpEff annotation
- Co-infection analysis
- Pan-virome landscape

→ variants + quant tables

MODULE 4

Phylogenetics

virome_phylo_pipeline

- Recombination (RDP5)
- Selection (HyPhy)
- Population genetics
- BEAST dating · MCC
- Transmission maps

→ trees + dynamics figs

MODULE 5

Data Submission

virome_submission_pipeline

- Topology triage
- MIUVIG metadata
- Sequin / suvtk
- tbl2asn .sqn build
- Hypothetical protein QA

→ submission-ready .sqn

SHARED INFRASTRUCTURE

Tool Library

30+ wrapped tools: Kraken2 · Bowtie2 · snpEff
VirHunter · ViReMa · BEAST · RDP5

Reference Databases

ICTV VMR taxonomy · plant-virus
reference DB · host genomes

Metadata Governance

Time/location parsing, normalization
& three-channel split

Reproducible Environment

pixi-pinned environments · stage
checkpoints · one-command reruns